OSNH52WAVDAPI

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DAT-300 – Data Validation: Getting the Right Data

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OS-52

I feel I should begin by explaining the title, ***OSNH52WAVDAPI***. It stands for **Open-Source New Hampshire 52 With a View Data API**, but neither roll off the tongue, so we'll call it OS-52 for short. This is where we arrive in this story: the future state. I'll explain the where-we-were—the before state—in a moment, but the true adventure was and is in the journey, from there to here. It is a perilous one, over hill and dell of data exploration and paintstaking validation, climbing mountainous peaks of varied information sources and traversing deep crevasses of conflicting data, arriving at the peak of understanding with an eagle's view of the wider landscape of what's genuinely important and useful information. Insightful. *Actionable.*

It is important to acknowledge the fact that AI Sherpas are doing most of the heavy lifting. Much—most[1]—of the back end code of OS-52 was generated by Claude Opus 5.0 (Anthropic, 2026), while I directed whic, ah architectures and code frameworks to use, and supplied the data sources. *This* document was/is completely human-written and largely off the cuff. But, before anything else, we should begin at the beginning, in beautiful scenic Southern New Hampshire.

Southern New Hampshire is beautiful in the late summer. New Hampshire has well over a hundred mountain peaks with summit trails. Many of them are challenging climbs over 4,000 feet. Many of them are fair climbs with varying degrees of challenge, but no particularly good view. That is why a group of New Hampshire senior hikers got together and issued the official list of New Hampshire's 52 With A View (Over the Hill Hikers). It is a curated list of 52 mountain peaks in New Hampshire between 2500 and 4000 feet summit height and spectacular views of the surrounding terrain. That was in 1990, and they have been actively curating and maintaining the list ever since, updating it as recently as 2025 but—perhaps archaically—maintaining it on a blogspot.

The list is popular, with other hiker community websites doing reviews of all of the mountains, where to park, which trails to take. The generated tons of data on trail lengths, gain in height to the summit, challenge level, quality of view at the top and along the way. Two great ones are New Hampshire Family Hikes and New England Waterfalls, which I will cite in the

[1] all

references. We also have geological and geographic data from sources, including the US Geological Survey and UNH's GRANIT, a detailed geospacial database. There is another dozen smaller supporting data sources of various types.

We certainly have plenty of data—both qualitative and quantitative. But it is disparate and disagreeing. Even the complete list has different mountains on different websites. And much of the qualitative data is subjective—where one blogger writes the difficulty level as challenging, the other rates as medium. One mountain on the border with Maine has multiple trail heads with one in each state. Location data varies. Names are abbreviated in one dataset but not another, e.g. *Mountain* versus *Mtn.* Rich data is presented in rigid HTML tables, but the truth is too multi-dimensional.

So that is our "current" state: rich, messy data, difficult to navigate, and conflicting. So where would like to be? I'd like to quickly search and filter the mountains based on criteria, and I want to see the data and details from various sources all in one place. And, to share the data as an open source with the rest of the community—New Hampshire hikers who love beautiful panoramas and don't need the challenge of New Hampshire's 48 over 4,000'. The user interface and experience—UI/UX—must be effortless as it filters and drills down on challenge rating versus summit height. It needs to let the user keep a checklist and maintain it online. And it needs to be free to host.

The Gap

Let us consider a metaphor. We, the adventurers, must cross a wide glacial crevasse to achieve our summit. We have three short climbing ladders which will just barely bridge the gap if we extend them all the way and tie them with a good climbing rope. The ladders are labeled E, T, and L. And as we extend them, we see the words *Extract*, *Transform*, and *Load*. They represent the processing of data integration for business intelligence. At such great heights, one or more sherpas is a must, which can represent AI assistants like Claude (Anthropic). The rope that keeps the process-ladders secure and connected is the tech stack—the software and services used.

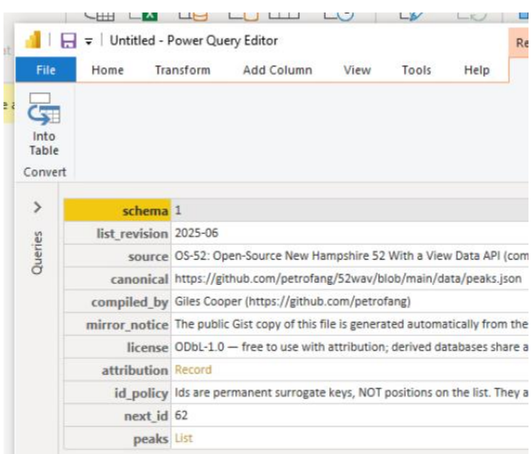
For us, one of the requirements was zero cost. This made GitHub pages an ideal choice, because the data can be freely shared under an open-source license. And GitHub Pages serves the page for free. The best sherpas aren’t free, but the rest will take you quite a way, if you know your basic mountaineering—you know common development tech stack, your tools, languages, libraries, etc. Any development environment like VS Code or even just terminal-based coding works. Knowing the libraries and tools, and languages like Python and JavaScript. And how to communicate with those AI sherpas—these are the twine that make the rope that is our tech stack, with which we bind the ladders of the ETL pipeline and traverse this deep data crevasse.

The first ladder is *Extraction*. We need to identify the data sources and assess their contents and the shape of the data. We find two hiker sites and two technical/official record open data sources. Since these are largely legacy hand-written tables, it’s out of scope to use something like this in Power BI—legacy tables in html documents on websites are challenging to scrape for data without a python library like Beautiful Soup (Richardson). We are fortunate enough to have the fully-realized future state already in our hands, and it exposes the true core of OS-52: the API.

It is a simple, read-only data warehouse of all available data, in a JSON NoSQL format. See link to the raw file in the references (Cooper). Power BI can draw on our already ETL’d data for further analysis, as seen loading in Figure 1.

Figure 1

Power BI Power Query Editor



Note. Metadata and notes are included in this NoSQL non-relational, hierarchical, non-tabular format of Java Script Object Notation, known as JSON.

Now that I’ve connected Power BI to it, I see this is going to be a pain to navigate the JSON. We have failed the Load function. But Suffice to say that in any format, our database represents a completed Extraction step, the first ladder over the gap. It also transforms the data.

But that transformation of data was a complex process, assisted by AI, to aggregate often subjective data between diverse sources. I investigate and adjudicate data disagreements. I identify which sources were the most active and thus have the highest quality of data in terms of timeliness. I decide what to include and what to leave behind—copyrighted material or low-quality data. I give full credit to all data sources. Federal and State data need reconciliation. The exact correct GPS locations are found, with web links for driving directions to the parking places, pull-offs and trailheads—not the summit, or the closest mailing address.

A newer-to-the-list mountain or two has not been reviewed by some sources, so we impute a median view rating of 8/10. Difficulty ratings are imputed based on the trail distance and rise in height vs the average of hikes with those stats. Retired hikes are included in the list, but a column was added to mark each as current or retired. Interestingly, one mountain was retired in 2020 because trees grew and it no longer had a view, but by 2025 it was re-added because they had done some tree work (New England Waterfalls).

So, that's the reach of the ladder marked Transform. Python was the right choice for these data sets, especially for the extract and transform steps—pending the CSV output. Power BI's Power Query can load and parse that easily. Where it shines will be in the visuals and reports it can provide. This brings to our third ladder: Load. The end is in sight—a future state with a catalog of fully aggregated, up-to-date, accurate and safe data and information about New Hampshire's beautiful 52 With a View.

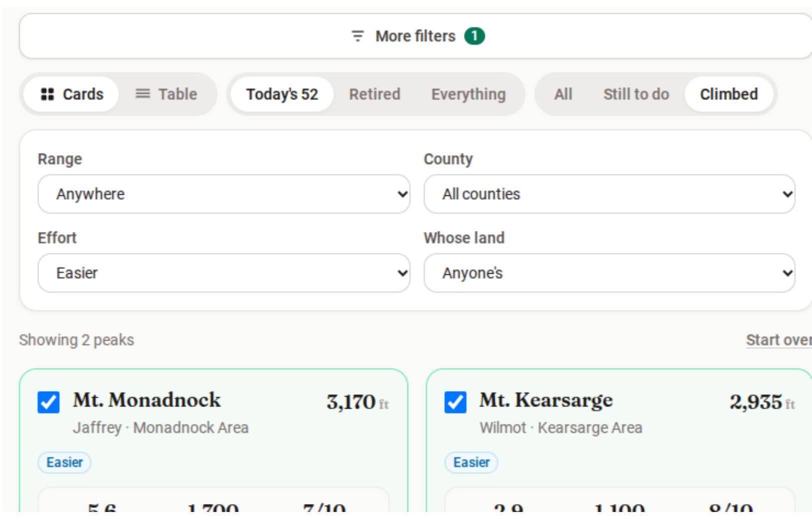
JSON isn't a neat tabular format, but it is hierarchical, which translates well into a web API when hosted on GitHub Pages. Once the data is collected, it's easy enough to deposit its final agreed state into the database file. It's essentially a data warehouse, but the data is simple, so it's easy enough to just download a single .json file. This is a deeply cleaned and useful data set for any other apps or websites about hiking in New Hampshire. But a JSON file is not useful to the user.

This is why it needed a front end or public access point. If you're going to look at it before the end, now is the time: https://petrofang.github.io/52wav/ The user interface and the whole user experience were designed to be user-friendly and follow the best practices and principles in modern user experience design. We have card and table views, basic filters for retired or completed climbs, and additional filters for geographical data and difficulty ratings. See Figure 2.

Now that we've crossed the third and final ladder—Load—in our ETL bridge over the chasm of conflicting data and disparate data sources, we approach the apex of usability and actionable insight. We now deliver the highest-quality data in the most open and accessible formats, on the most solid and timely sources, packed with a friendly and easy user interface. Our poetic mountain of data does, too, yield a fantastic view from its apex. We have a clear view of literally the entire landscape and terrain of New Hampshire. We see every parking space, pull-off, trailhead and summit. We know here to find the easy ones, the hard ones, the best-viewing ones. You check them off as you hike them. The product is complete. The data is made public. This is the completed Open-Source New Hampshire 52 With a View Data API.

Figure 2

Filters in the OS-52 Front End



Description. Primary and additional filters are applied to filter for mountains already climbed with an easier effort rating under any class of land-owner.

There remain a couple of questions about data ethics and regulations. We must follow respectful data policy in our extraction. The data includes detailed written trail logs and leg work by individuals. In those cases, we use that information in our decision making of the transformation process, but do not include it in the final dataset. Human writing is real intellectual property, and I wouldn't offend that pretending to assume that anything written on a blog in the 90s was intended as public material issued without license for automated consumption.

Legal and jurisdictional questions were also asked and resolved, such as proper federally recognized names of each peak—Mt. Monadnock is federally "Monadnock Mountain," roughly translated from Abenaki as "Lonely Mountain Mountain." Some summits were in different towns, counties and even states than the trailhead. Eagle Crag's summit is in New Hampshire, but the preferred trailhead is in Maine (New England Waterfalls). In these cases, we used UNH's GRANIT data to determine the town, county, and landowner of the summit with GPS (University of New Hampshire) and maintain the parking area's driving address as separate data entities.

With a proper tabular dataset, we may load it into Power BI and report and analyze it in various ways with reports and dashboards with detailed and colorful visuals. But I will leave that for the next gap analysis.

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